

Script-Controlled ACL – Restrict Record Access Based on Field Value

Naan Mudhalvan / Skill Wallet Project - Design Phase Document

1. Project & Team Information

- Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value
- Phase: Design Phase (System Architecture & Technical Design)
- Team ID: SWTID-2026-9611
- Team Leader: Adhsa Rani P
- Team Members: Gayathri A, Parkavi P, Priyadharshini M, Chinthanika P
- Platform: ServiceNow / SkillWallet (SmartBridge)

2. System Architecture & High-Level Design

- Overview: The system uses ServiceNow's Access Control List (ACL) framework to govern record access. Security decisions are processed sequentially:
 1. Role Verification (User possesses required custom role).
 2. Condition Evaluation (Static UI builder checks).
 3. Script Execution (Contextual JavaScript runtime evaluation).
- Data Model Target:
 - Table: `u_institution_details`
 - Controlled Fields: `u_branch`, `state`, `is_confidential`
 - Access Operations: READ, WRITE, CREATE, DELETE

3. Technical Specifications & Script Configuration

3.1 ACL Rule: Delete Operation Restriction

- Type: Record
- Operation: Delete
- Name / Table: `u_institution_details`
- Active: true
- Roles Required: `bb4`
- Script Logic:

JavaScript

```
(function executeRule(current, previous) {
```

```

    if (!gs.hasRole('bb4')) {
        return false;
    }
    if (current.u_branch == 'EEE') {
        return true;
    }
    return false;
})();

```

4. Control Flow & Evaluation Pipeline

1. Incoming request for record action on u_institution_details.
2. Role Check: User has required custom role?
3. Static Filter Check: UI conditions matched?
4. Script Evaluation: Field criteria evaluated dynamically (current.u_branch == 'EEE').
5. Decision: Grant or Deny access.

5. Test Case Design Matrix

Test Case ID	Target Operation	User Role	Record Field Value	Expected Decision
TC-DES-01	DELETE	bb4	u_branch == 'EEE'	ALLOW
TC-DES-02	DELETE	bb4	u_branch == 'CSE'	DENY
TC-DES-03	DELETE	bb3	u_branch == 'EEE'	DENY
TC-DES-04	WRITE	bb3	state == 'Active'	ALLOW
TC-DES-05	WRITE	bb3	state == 'Closed'	DENY